Product Health and Insights Summary

Part Identification Repository API | Thematic synthesis for executive review

Total issues	Critical	High	Low
12	4	7	1

Technical stability and backend performance

Single-request design degrades under repeated, automated load.

N+1 query degradation	Critical
Thread contention and locking	High
Memory growth on large payloads	High

Developer experience and workflow efficiency

Batch jobs run through a single-lookup interface.

Manual looping exhaustion	Critical
Rate limiting and throttling	High
Brittle error recovery	High

Payload efficiency and AI cost

Frontend-optimized responses inflate token and compute cost.

Payload bloat and token waste	Critical
No dynamic compression	High

Security and data governance

No sanctioned bulk path has created shadow workarounds.

Shadow workarounds	Critical
Loss of risk-assessment visibility	High

Data integrity and consistency

High-frequency access undermines cache reliability.

Stale or out-of-sync cache data	High

Minor technical debt

Aggregated low-priority items.

Wrapper maintenance, field ordering, per-item error reporting	Low

Severity scale: Critical, High, Medium, Low. Counts reflect the 12 line items in the thematic synthesis.